

Literature Survey on Vulnerability Assessment and Penetration Testing (VAPT)

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Date of Submission:

5th October 2025

Academic Year: 2025–2026

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Abstract

In this survey, we examine VAPT and how it is used in various technological environments. We also compare it with the different stages of cyber attack lifecycles and discuss the best strategies, major difficulties, and directions for future research.

Introduction

To protect digital assets from modern cyber threats, VAPT plays a crucial role. It has a mixture of automated scans with hands-on testing to reveal actual business risks and improve security overall.

Defining VAPT

1. **Vulnerability Assessment:** Systematic discovery and prioritization of security weaknesses using automated tools and manual testing. Focuses on a broad coverage but does not exploit issues.
2. **Penetration Testing:** Ethical hackers run realistic attack simulations to see how vulnerabilities can actually be exploited

and what damage they might cause. The results give us clear guidance on how to fix the issues.

3. **Integrated Approach:** A combined VAPT process delivers both breadth and depth, addressing the weaknesses of either method alone.
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VAPT Methodology

1. **Pre-Engagement:** Establish boundaries, getting proper authorisations and decide on methodologies before testing begins.
 2. **Reconnaissance:** Gather intelligence about the target through active and passive reconnaissance.
 3. **Vulnerability Analysis:** Use tools and hands-on testing to uncover vulnerabilities.
 4. **Exploitation:** Attempt controlled attacks to validate risks. Do not collect or store the data you found after exploitation.
 5. **Reporting:** Summarize results, assign risk ratings, and provide guidance for fixing issues.
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Cyber Attack Lifecycle vs VAPT

Frameworks like Cyber Kill Chain and Mitre Att&ck describe how attackers move through each stage of an attack, while VAPT focuses on testing defenses to see how well they stand up against those tactics.

	Cyber Attack Lifecycle	VAPT Methodology
Purpose	Model attacker strategies	Evaluate defenses
Constraints	Few, illegal actions possible	Strict ethical/legal limit
Scope	Expands dynamically	Fixed, pre-agreed
Reporting	Minimal	Extensive business/tech detail

VAPT Implementation Environments

1. **Network Infrastructure:** Focuses on devices, wireless, and remote access. Balances security with service availability.
 2. **Web Applications:** Prioritizes input validation, authentication, logic flaws, and client-side attacks.
 3. **Cloud:** Addresses about dividing responsibilities, sharing the same system with others, setting up configurations, using APIs safely, and staying compliant with regulations.
 4. **IoT & Mobile:** Highlights default credentials, insecure services, firmware and hardware flaws, and mobile OS-specific issues.
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Similarities and Differences Across Environments

Similarities:

- Universal phased approach: planning, reconnaissance, assessment, exploitation, reporting.
- Standardized risk scoring (CVSS).

Differences:

- The tools we use depend on the environment—for example, wireless analyzers for networks, code reviews for web apps, and firmware checks for IoT devices.
 - Different types of assets need different safety rules and risk checks.
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VAPT Benefits and Limitations

Benefits:

- VAPT identifies real-world threats.
- Proves compliance with security regulations.
- Guides smart spending on protection.
- Improves security awareness among employees.

Limitations:

- It is not ongoing—risks can appear after the test.
- The outcome depends on who carries it out and what resources they have.

- Tests could disrupt systems unless managed carefully. And because of legal and ethical limits, it can not go as far as an actual hacker might.
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Current Trends and Research

- **Automation:** Let AI handle the scanning so you spend less time on manual checks and find more problems with higher accuracy.
 - **Cloud-Native VAPT:** Cloud-native security testing is changing with serverless and microservices, so you need a fresh toolkit.
 - **IoT/Edge:** Scaling and adapting VAPT approaches to handle mass, heterogeneous deployments with specialized protocol analysis.
 - **Integration:** Red Team exercises and attack simulations for defence validation using attack lifecycle knowledge.
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Conclusion

- VAPT helps organizations find weak spots in their systems and see how those can be attacked in real life.
- VAPT and cyber attack lifecycles are different, but together they make security stronger. Lifecycles show what attackers do, and VAPT checks if defenses really work.
- This approach works for all kinds of environments—including regular networks, cloud systems, and smart devices—by using the same basic ideas and adjusting them to each situation.
- As technology changes, VAPT is also changing with more automation and tools made for cloud and smart device security.
- Organizations should stick to proven methods but stay open to new ways to keep their systems safe.

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Annexure: Report Templates for Upcoming Phases

Phase 2: Reconnaissance Phase Report Template

Summary

- Reconnaissance scope and objectives
- Information gathering
- Key intelligence findings
- Identified attack vectors and entry points

Technical Findings

- Target system architecture
- Network topology and service enumeration
- Application mapping and functionality analysis
- Identified personnel and organizational structure

Intelligence Assessment

- Information sensitivity analysis
- Exposure risk evaluation
- Attack surface quantification
- Threat landscape contextualization

Phase 3: Vulnerability Assessment Report Template

Summary

- Assessment scope and coverage statistics
- Vulnerability distribution by severity
- Critical findings requiring immediate attention

- Overall security posture evaluation

Vulnerability Catalog

- Detailed vulnerability descriptions
- CVSS scoring and risk prioritization
- Affected system components and services
- Exploitability assessment and impact analysis

Remediation Roadmap

- Priority-based remediation timeline
- Resource requirements and cost estimates
- Patch management recommendations
- Security control enhancement suggestions

Phase 4: Penetration Testing Report Template

Summary

- Testing objectives and success criteria
- Exploitation results and access gained
- Business impact demonstration
- Critical security gaps identified

Attack Narrative

- Step-by-step exploitation methodology
- Tools and techniques utilized
- Privilege escalation and lateral movement
- Data access and exfiltration capabilities

Security Control Assessment

- Detective control effectiveness evaluation
- Preventive measure bypass techniques
- Incident response capability testing
- Recovery and resilience assessment